may be used as tools for the study of plant taxonomy. In the case of animal studies, the classical tools are Museum, Taxonomical Keys and Zoological and Marine parks.

The important components of the taxonomical tools are field visits, survey, identification, classification, preservation and documentation. Many tools are being used for taxonomical studies, amongst them some of the important tools are discussed below:

Arignar Anna Zoological Park, also known as the Vandalur Zoo is in the south western part of Chennai, Tamil Nadu, spreads over an area of 1500 acres, is one of the largest zoological parks in India. The zoo houses 2,553 species of both flora and fauna.

The classical taxonomical tools

Taxonomical Keys: Keys are based on comparative analysis of the similarities and dissimilarities of organisms. There are separate keys for different taxonomic categories.

Museum: Biological museums have collection of preserved plants and animals for study and ready reference. Specimens of both extinct and living organisms can be studied.

Zoological parks: These are places where wild animals are kept in protected environments under human care. It enables us to study their food habits and behaviour.

Marine parks: Marine organisms are maintained in protected enviroments.

Printed taxonomical tools consist of identification cards, description, field guides and manuals.

Molecular taxonomical tools

Technological advancement has helped to evolve molecular taxonomical tools from classical tools to molecular tools. The accuracy and authenticity is more significant in the molecular tools. The following methods are being used for taxonomical classification.

Molecular techniques and approaches such as **DNA barcoding** (short genetic marker in an organism's **DNA** to identify it as belonging to a particular species), **DNA hybridization** (measures the degree of genetic similarity between pools of DNA sequences), **DNA fingerprinting** (to identify an individual from a sample of DNA by looking at unique patterns in their DNA), Restriction Fragment Length Polymorphisms (RFLP) analysis (difference in homologous DNA sequences that can be detected by the presence of fragments of different lengths after digestion of the DNA samples), and Polymerase Chain Reaction (PCR) sequencing (to amplify a specific gene, or portion of gene,) are used as taxonomical tools.

Automated species identification tools

It consists of Cyber tools. For example: ALIS, DAISY, ABIS, SPIDA, Draw wing, etc.

ALIS ➜ Automated Leafhopper Identification System.

DAISY ➜ Digital Automated Identification System.

ABIS ➜ Automatic Bee Identification System.

SPIDA ➜ Species Identified Automatically (spiders, wasp and bee wing characters).

Draw wing ➜ Honey bee wing identification.

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